

The Locally Nilpotent Part of Nonboundary Localization Quotients

Corrected E-action and primitive recursion; the weight-filtration claims are withdrawn

Progress report, corrected after the audit of 2026-09-22

September 22, 2026

Abstract

This is a corrected version of the September 22 note on $G_c = \{v \in \mathcal{T}_{f_c} M_{a,b} : e_{-c}^n v = 0 \text{ for some } n\}$ in the deep region. The independent audit ([audits/2026-09-22/](#)) found explicit counterexamples to the E-action formula, to the primitive-space description $P_\nu \cong \{X : T^{\nu+1}(X)u = 0\}$, to the claim $P_0 = \mathbb{C}w_1$ at $\mu = -2$, and to the statement that the weight tails $G^{\geq \nu}$ are affine submodules. This version records the corrected E-action through the full components A_r , the corrected triangular recursion for primitive vectors, the infinite family $Z^n w_1$ in P_0 , the complete set of projections including the lowering map $\ell_j : P_\nu \rightarrow P_{\nu-2}$, and withdraws the weight-filtration consequences. The statements proved earlier and not affected are the boundary isomorphism, the deep-region simplicity criterion, the unique maximal submodule with simple head, and the algebraic central reduction compatibility. The internal classification of G_c (simplicity, socle, length) is open.

1 Definitions, notation, and assumptions

Affine algebra and induced modules. As in the unified note: $\widehat{\mathfrak{sl}}_2 = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K$,

$$[e_i, f_j] = h_{i+j} + i\delta_{i+j,0}K, \quad [h_i, h_j] = 2i\delta_{i+j,0}K, \quad [h_i, e_j] = 2e_{i+j}, \quad [h_i, f_j] = -2f_{i+j},$$

K central, $[e_i, e_j] = [f_i, f_j] = 0$. Fix $a, b \in \mathbb{Z}$, $a + b \geq 0$, $L = a + b + 1 \geq 1$, and $S_{a,b} = \text{span}_{\mathbb{C}}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}$. The character $\chi : S_{a,b} \rightarrow \mathbb{C}$ is determined by $\kappa = \chi(K)$ and $\lambda_i = \chi(h_i)$, $0 \leq i \leq L$, with $\lambda_i = 0$ for $i > L$; $M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_\chi$, $u = 1 \otimes 1_\chi$. The central level is denoted by κ (not c), reserving c for the localization index.

Deep-region localization. Fix $c < b$ with $c \leq -a - 1$ (equivalently $d_* := b - c \geq L$) and set $F = f_c$. The spectral-flow twist σ_{-c} identifies $\mathcal{T}_F M_{a,b}(\chi)$ with $Q = \mathcal{T}_{f_0} M_{A,B}(\chi')$, where $A = a + c \leq -1$, $B = d_* \geq L$, and $\mu := \chi'(h_0) = \lambda_0 - c\kappa$. Write $E = e_0$, $H = h_0$, $w_m = f_0^{-m}u + M$. Then $[E, f_0] = H$, $Eu = 0$, $Hu = \mu u$.

PBW notation.

$$\mathcal{X} = \left\{ X = f_{j_1} \cdots f_{j_p} e_{i_1} \cdots e_{i_q} h_{r_1} \cdots h_{r_s} : j_t \leq B, j_t \neq 0, i_t \leq A, r_u < 0 \right\},$$

including the empty word. $\{F^{-m}Xu : m \geq 1, X \in \mathcal{X}\}$ is a basis of Q . Write $\deg X = 2q - 2p$; then $H(F^{-m}Xu) = (\mu + 2m + \deg X)F^{-m}Xu$. Let $V_0 \subset M$ be the span of $\{Xu : X \in \mathcal{X}\}$ (the F -free part), so that $M = \bigoplus_{r \geq 0} f_0^r V_0$.

The locally nilpotent part. $G := \{v \in Q : E^n v = 0 \text{ for some } n \geq 1\}$ is a $\widehat{\mathfrak{sl}}_2$ -submodule, and every nonzero $\widehat{\mathfrak{sl}}_2$ -submodule of G contains a nonzero vector of $P_\nu := \ker E \cap G \cap Q_\nu$ (apply powers of E and take the last nonzero image). In the shallow region $-a \leq c < b$, $E = e_{-c}$ is injective on Q , so $G = 0$; only the deep region carries a nonzero locally nilpotent part.

2 The corrected E-action

Lemma 1. For $x \in V_0$ define linear maps $A_r : V_0 \rightarrow V_0$ by

$$Ex = \sum_{r \geq 0} f_0^r A_r x,$$

a finite sum for each x , with $A_0 = \pi_0 \circ \text{ad}(e_0)$ the old operator T . If $Hx = (\mu + \delta)x$ and $m \geq 1$, then in Q ,

$$E(f_0^{-m}x) = \sum_{r=0}^{m-1} f_0^{-(m-r)} A_r x - m(\mu + \delta + m + 1)f_0^{-m-1}x. \quad (1)$$

Proof. By the triple identity $Ef_0^{-m} = f_0^{-m}E - mf_0^{-m-1}H - m(m+1)f_0^{-m-2}f_0$ and $Ex = \sum_r f_0^r A_r x$; the term $f_0^{-m}f_0^r A_r x$ survives in Q for $0 \leq r < m$ as $f_0^{-(m-r)} A_r x$, and vanishes for $r \geq m$. $\square$

Corollary 2. The earlier formula retaining only A_0 is false. For $X = f_{-1}f_1$ one has $EXu = (f_1h_{-1} + \lambda_1f_{-1} - 2f_0)u$, hence

$$E(f_0^{-2}f_{-1}f_1u) = f_0^{-2}(f_1h_{-1} + \lambda_1f_{-1})u - 2w_1 - 2(\mu - 1)f_0^{-3}f_{-1}f_1u,$$

and the A_0 -only formula misses the nonzero term $-2w_1$ (machine-checked in `work/check_audit_regressions.py`).

3 The primitive space

Lemma 3. If $0 \neq v \in Q_\nu$ and $Ev = 0$, then $\nu \in \mathbb{Z}_{\geq 0}$, and the highest pole of v is exactly $\nu + 1$.

Proof. Write $v = \sum_{m=1}^N f_0^{-m}x_m$ with $x_m \in V_0$. At pole $N+1$ only the diagonal of (1) survives, forcing $N = \nu + 1$. $\square$

Theorem 4. A nonzero vector $v = \sum_{m=1}^{\nu+1} f_0^{-m}x_m \in Q_\nu$ is primitive if and only if its layers satisfy the triangular system

$$\sum_{m=k}^{\nu+1} A_{m-k}x_m = (k-1)(\nu - k + 2)x_{k-1}, \quad 1 \leq k \leq \nu + 1, \quad x_0 = 0, \quad (2)$$

and the bottom condition at $k = 1$ is the compatibility $\sum_{m=1}^{\nu+1} A_{m-1}x_m = 0$. The primitive space P_ν is the linear space of solutions of this system; it is not in general a set of single monomials, and the old description $P_\nu \cong \{X : T^{\nu+1}(X)u = 0\}$ is false.

Proof. Equate coefficients of f_0^{-k} in $Ev = 0$ using (1). For $k = \nu + 1$ the right-hand side is $\nu x_\nu = A_0x_{\nu+1}$, so the top layers are determined downward; the $k = 1$ equation is the remaining compatibility condition. For $\nu = 1$ the system reads $x_1 = A_0x_2$ and $(A_0^2 + A_1)x_2 = 0$. $\square$

Corollary 5. The old criterion genuinely fails. The element

$$Z = h_{-1}^2 + 4f_{-1}e_{-1} + 2h_{-2} = h_{-1}^2 + 2(e_{-1}f_{-1} + f_{-1}e_{-1})$$

commutes with E , f_0 , and H (it need not be central in $\widehat{\mathfrak{sl}}_2$). At $\mu = -1$, the vector $p = \lambda_1f_0^{-1}u + f_0^{-2}f_1u$ lies in P_1 , and $Zp \in P_1$ has layers $x_1 = (\lambda_1Z - 4h_{-1})u$, $x_2 = (f_1Z - 4(\kappa+1)f_{-1})u$ with $A_0^2x_2 = 8e_{-1}u \neq 0$; the missing term $A_1x_2 = -8e_{-1}u$ restores the compatibility (machine-checked).

Corollary 6. At $\mu = -2$, $\dim P_0 = \infty$: the vectors $Z^n w_1 = f_0^{-1}Z^n u + M$ ($n \geq 0$) are primitive and linearly independent, their leading PBW degrees being $2n$ with leading symbols $(h_{-1}^2 + 4f_{-1}e_{-1})^n \neq 0$. The earlier claim $P_0 = \mathbb{C}w_1$ is withdrawn.

4 The $\mathfrak{sl}_2$ -decomposition and the projections

Theorem 7. *Every vector of G lies in a finite-dimensional $\mathfrak{sl}_2(E, H, f_0)$ -submodule, and*

$$G \cong \bigoplus_{\nu \in \mathbb{Z}_{\geq 0}} P_\nu \otimes L(\nu)$$

as transverse $\mathfrak{sl}_2$ -modules, where $L(\nu)$ is the $(\nu + 1)$ -dimensional irreducible and P_ν is the multiplicity space. This decomposition is correct and is retained; it is not a decomposition into $\widehat{\mathfrak{sl}}_2$ -submodules.

Proof. E and f_0 are locally nilpotent on G , so each vector generates a finite-dimensional $\mathfrak{sl}_2$ -module; complete reducibility is Weyl's theorem, and primitive weights are nonnegative integers. $\square$

The principle behind all projections is that the current modes form the adjoint module $L(2)$ under the transverse $\mathfrak{sl}_2$, and $L(2) \otimes L(\nu) \cong L(\nu + 2) \oplus L(\nu) \oplus L(\nu - 2)$ for $\nu \geq 2$.

Proposition 8. *Let $p \in P_\nu$ have top layer X at pole $\nu + 1$.*

- (i) *For $i \leq A$: $e_i p \in P_{\nu+2}$, and its top layer is $-2f_i u$ for $\nu = 0$ (example: $e_{-1} w_1 = f_0^{-1} e_{-1} u - f_0^{-2} h_{-1} u - 2f_0^{-3} f_{-1} u$ at $\mu = -2$), and in general the top pole $\nu + 3$ carries the correction layer of $[e_i, f_0^{-m}]$.*
- (ii) *For $r < 0$ and $\nu \geq 1$: $h_r p + \frac{2}{\nu+2} f_0 e_r p \in P_\nu$, with top layer $\frac{\nu}{\nu+2}(-h_r X + \frac{1}{\nu} f_r A_0 X)$; for $\nu = 0$, $h_r p = -f_0 e_r p$.*
- (iii) *For $j \geq 1$: $E^2 f_j p = -2e_j p$, and $f_j p$ has a component in $P_{\nu+2}$ and one in P_ν .*
- (iv) *For $\nu \geq 2$ and all j : the lowering projection*

$$\ell_j(p) = f_j p - \frac{1}{\nu} f_0 h_j p - \frac{1}{\nu(\nu+1)} f_0^2 e_j p \in P_{\nu-2}, \quad (3)$$

the $L(\nu - 2)$ -component of the tensor product above. It is nonzero: at $\mu = -2$, for $p_2 = e_{-1} w_1 \in P_2$,

$$\ell_1(p_2) = f_0^{-1} \left(\frac{1}{3} f_1 e_{-1} + \frac{\lambda_1}{6} h_{-1} + \frac{2\kappa-2}{3} \right) u + M \in P_0 \setminus \{0\}.$$

Proof. (ii) and (iii) use $[E, h_r] = -2e_r$, $[f_0, e_r] = -h_r$ and $[E, f_j] = h_j$ as before; (iv) is verified by applying E to the right-hand side of (3) and using $E f_0^s = s(\nu - s + 1) f_0^{s-1}$ on the block of p . The nonvanishing example is machine-checked. $\square$

5 Submodule structure: what is withdrawn and what remains open

The weight tails are not affine submodules. The earlier note claimed that $G^{\geq \nu} := \bigoplus_{\nu' \geq \nu} P_{\nu'} \otimes L(\nu')$ is a $\widehat{\mathfrak{sl}}_2$ -submodule. This is false.

- (i) Central obstruction: at $\mu = -2$ and $\kappa \neq 0$, take $p = w_1 \in L(0)$. Then $h_{\pm 1} p = -f_0 e_{\pm 1} p$ lies in the blocks of top weight 2, so both $h_1 p$ and $h_{-1} p$ belong to $G^{\geq 2}$. Were $G^{\geq 2}$ a submodule, $[h_1, h_{-1}] p = 2\kappa p$ would belong to it, contradicting $p \in L(0)$ and $\kappa \neq 0$ (machine-checked).
- (ii) The lowering channel: $\ell_1(e_{-1} w_1) \in P_0 \setminus \{0\}$ at $\mu = -2$ shows that modes map P_2 into P_0 , so the tails are not stable even at $\kappa = 0$.

Consequently there are no nonzero trivial affine quotient modules of Q at level $\kappa \neq 0$: K acts by κ while all h -modes acting by zero would force $[h_1, h_{-1}] = 2K$ to act by zero.

Withdrawn claims. The following statements, derived in the earlier version from the invalid filtration, are withdrawn as proved results: that G is never simple; that $\text{soc}(G) = 0$ and $\text{soc}(Q) = 0$; that G has infinite composition length; that the quotients $G^{\geq \nu}/G^{\geq \nu+1}$ are simple and form the head. Their negations are not proved either: the internal classification of G_c is open.

What remains proved. The boundary isomorphism, the deep-region simplicity criterion $\mathcal{T}_{f_c} M_{a,b}$ simple $\iff \lambda_0 - c\kappa \notin \mathbb{Z}$ (under $\lambda_L \neq 0$, $\kappa \neq -2$), the unique maximal submodule G_c with simple head Q_c/G_c and nonsplit extension at integral parameters, the algebraic central reduction compatibility, the transverse $\mathfrak{sl}_2$ -decomposition of G_c , and the corrected recursion of Theorem 4 with the projections of Proposition 8. A correct submodule analysis of G_c must track all three channels $L(\nu + 2)$, $L(\nu)$, $L(\nu - 2)$ simultaneously; the old filtration is abandoned.

6 Verification status

The audit `audits/2026-09-22/` (report, independent checker with 154 exact checks, results, log) is part of this repository. Its four classes of counterexamples are reproduced by `work/check_audit_regressions.py` (24 checks): the E-action residual $-2w_1$; the truncated- T residual $-4e_{-2}u$; the Z -family in P_0 and the $\mu = -1$ corrected-recursion example; the obstruction $[h_1, h_{-1}]w_1 = 2\kappa w_1$; the nonzero $\ell_1(p_2)$. These are corroborations, not formal proofs, and the open status of the submodule classification should be reflected in any summary of this project.

References

- [1] V. Futorny, X. Guo, Y. Xue and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2 (2024), Theorem 1.1. <https://arxiv.org/abs/2404.03855v2>.
- [2] Independent audit of 2026-09-22, `audits/2026-09-22/audit_report_cn.md` and `audits/2026-09-22/independent_pbw_audit.py` (154 exact checks).
- [3] Project working notes: *Boundary and Nonboundary Localization Quotients of Smooth Affine Modules* (unified note, September 2026) and the addendum of September 15, 2026.